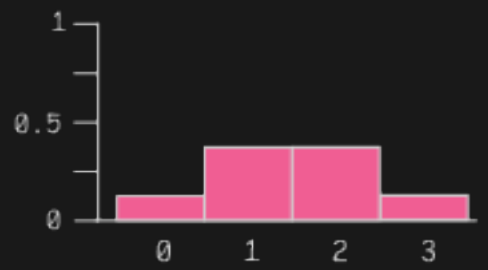
RANDOM VARIABLES

Probability experiment:
Flip a fair coin 3 times.

Bar chart of the outcomes
(often qualitative variable)

Histogram for
quantitative variable



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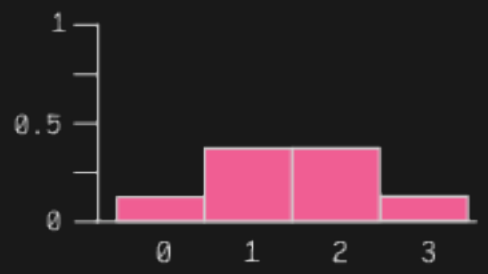

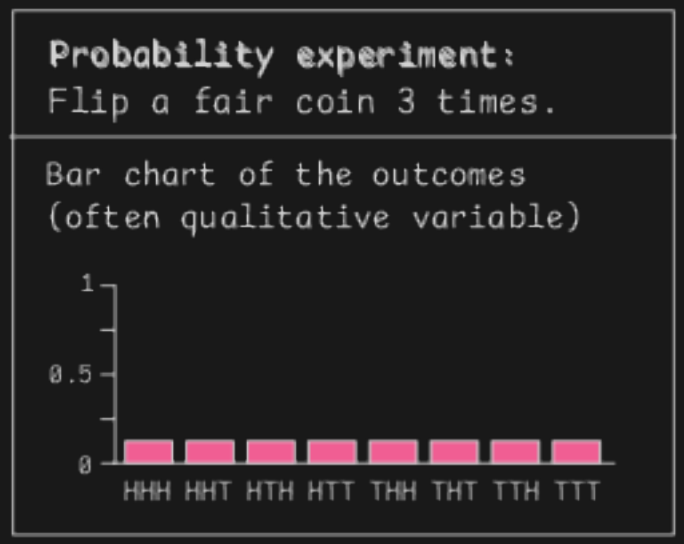

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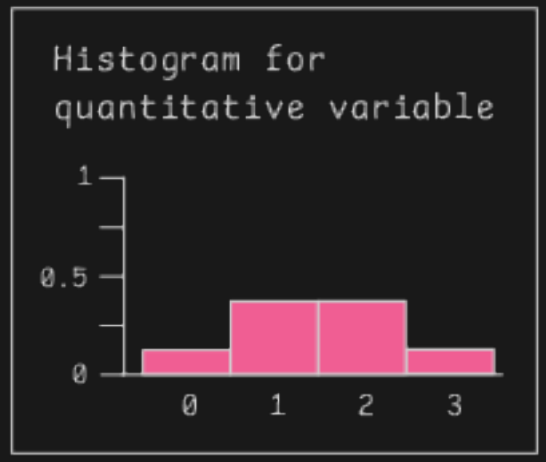

Formally, a random variable converts qualitative/ quantative variable to quantitative variable.

But we think of a random variable as the resulting quantitative variable.

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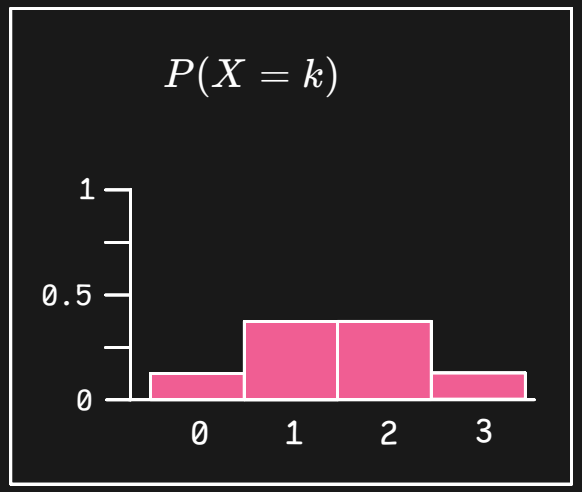

Probability distribution or probability mass function (PMF) of X

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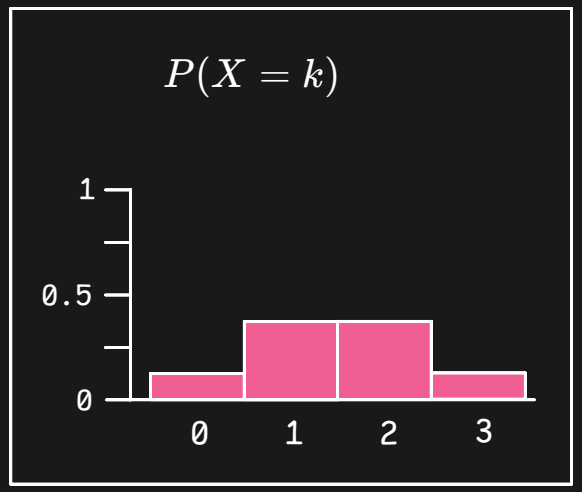

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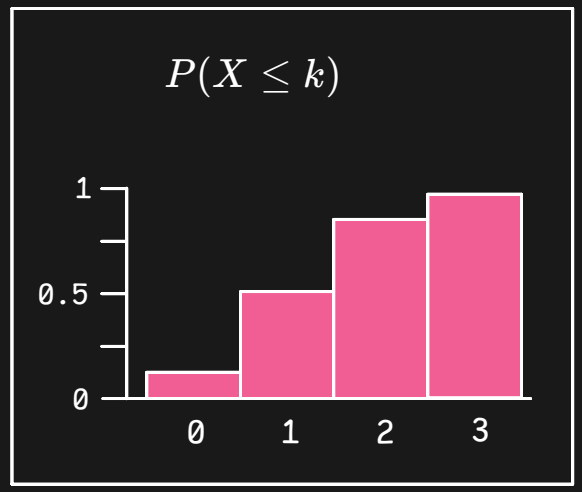

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General binomial RV X:

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$$\Rightarrow P(X = 2) = \binom{10}{2}(0.25)^2(0.75)^8$$

General geometric RV $Y \sim \mathrm{Geometric}(p)$:

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$$\Rightarrow P(Y = 4) = (5/6)^4 \cdot (1/6)$$

In-class exercise

Graph the PMF and CDF of the geometric random variable $Y \sim \mathrm{Geometric}(1/2)$ where probability p of heads (aka success) is $1/2$.

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But there are also continuous RVs.

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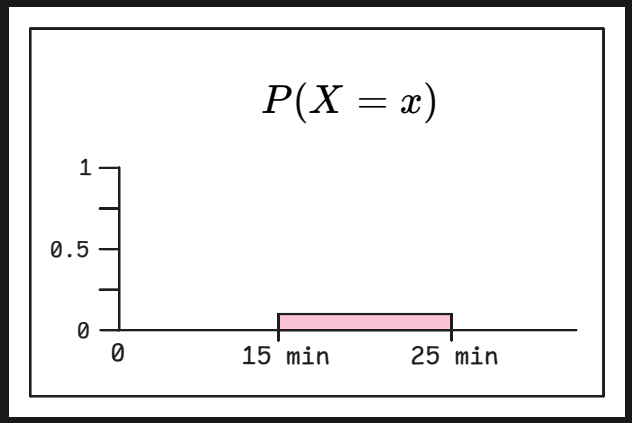
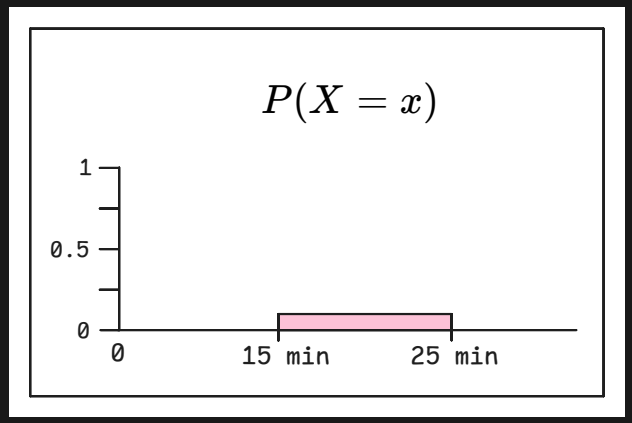


Probability density function (PDF) of X

Domino's Pizza is done at a time X that varies uniformly between 15 and 25 min.

PDF of X is constant with total area 1, so $f(x) = p(X = x) = 1/(25 - 15) = 0.1$ if $15 \leq x \leq 25$ and $f(x) = 0$ otherwise.



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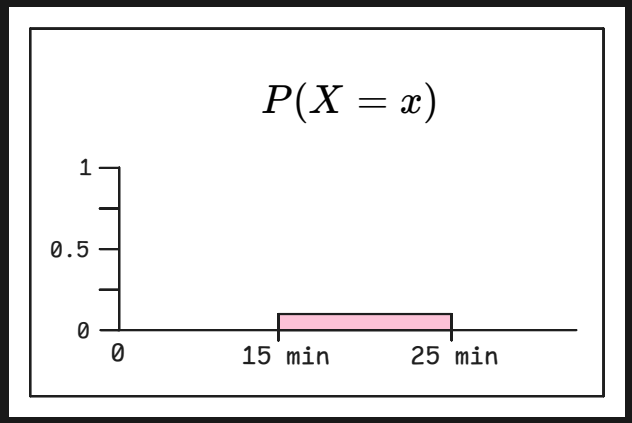

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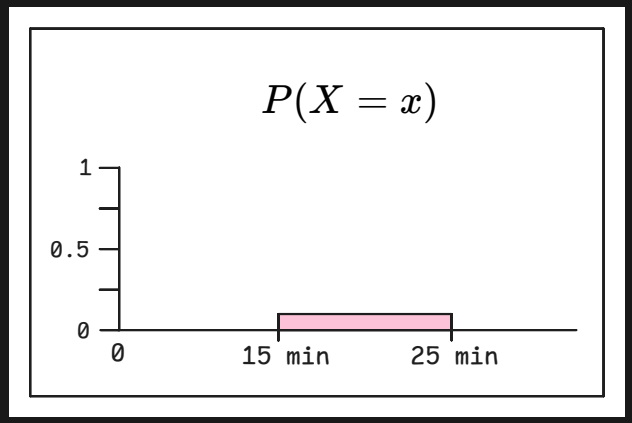

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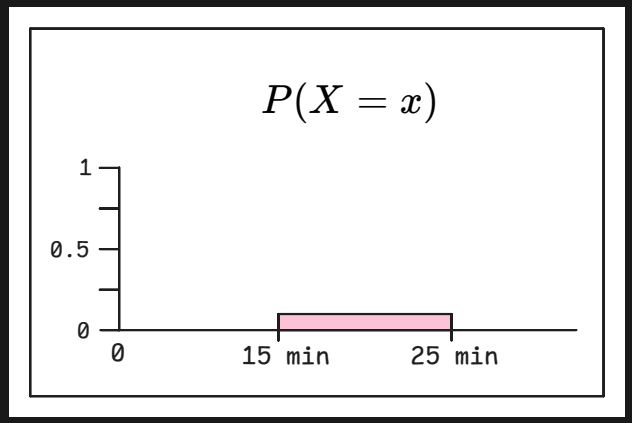

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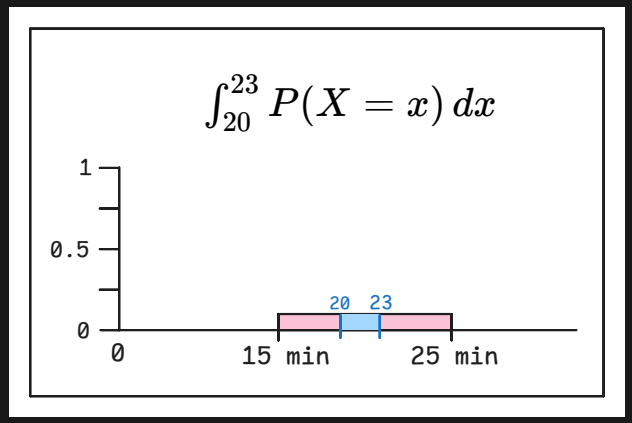

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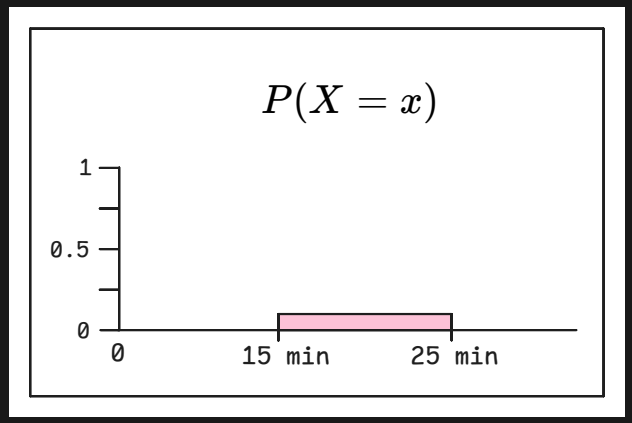

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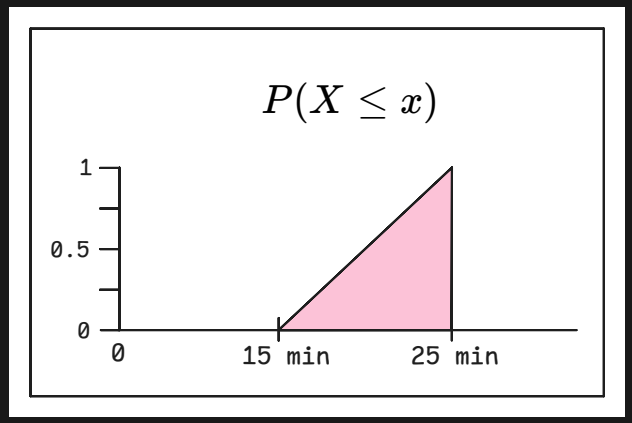

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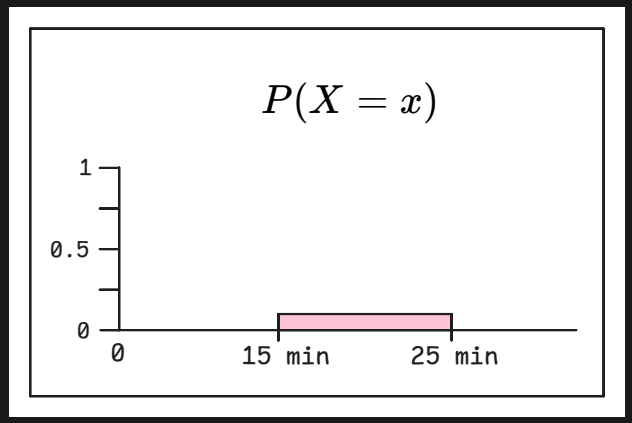

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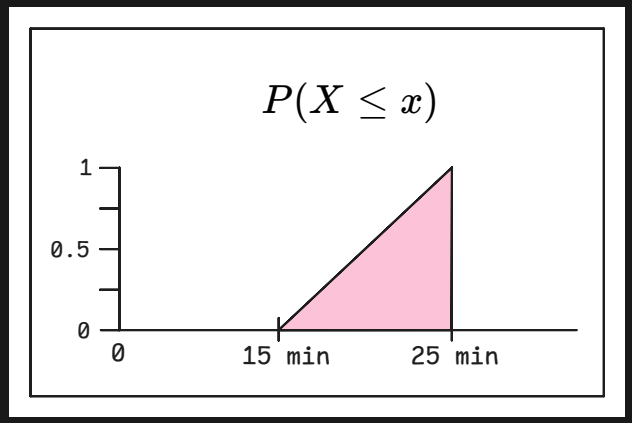

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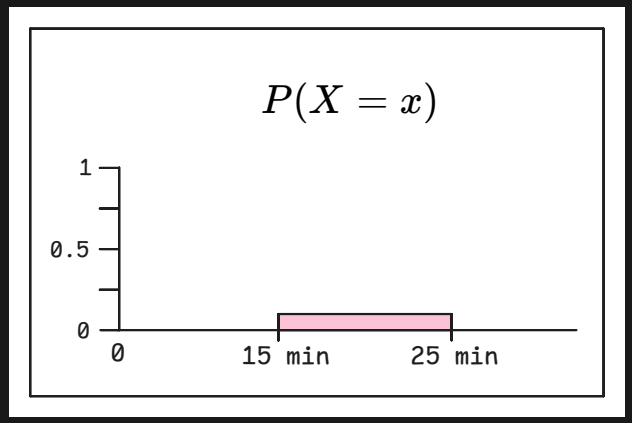

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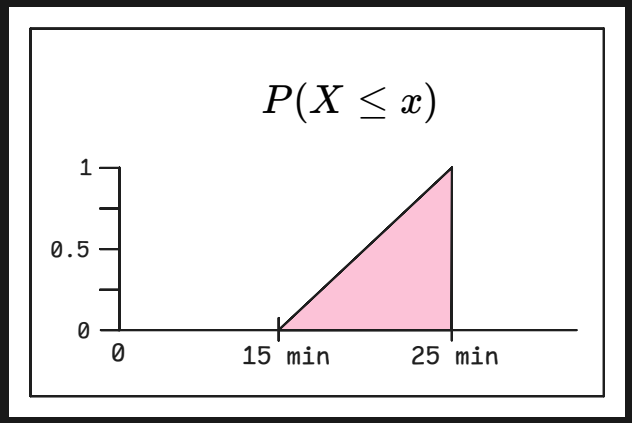

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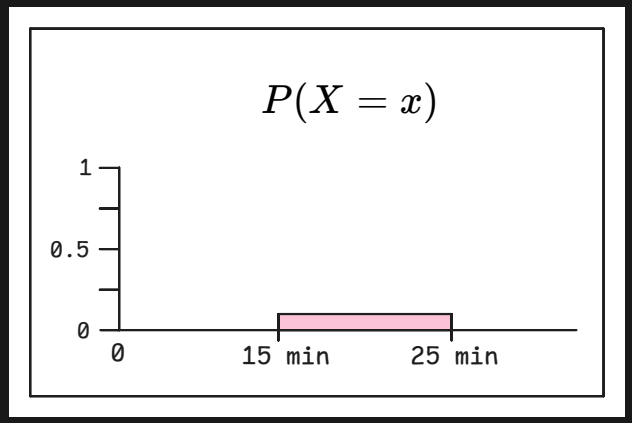
$$= \begin{cases} 0 & \text{if } x > 25 \\ 0.1(x - 15) & \text{if } 15 \leq x \leq 25 \\ 1 & \text{if } x < 15 \end{cases}$$



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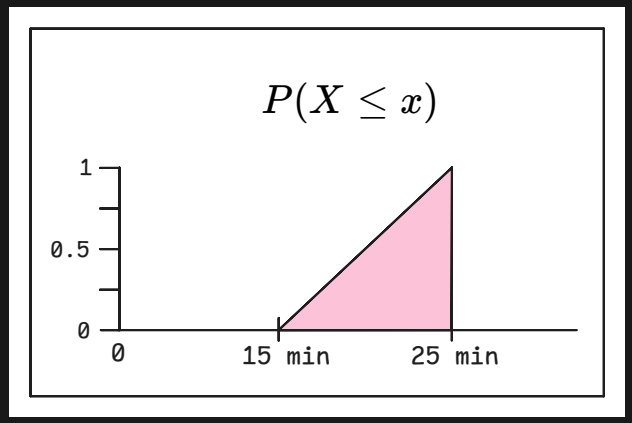

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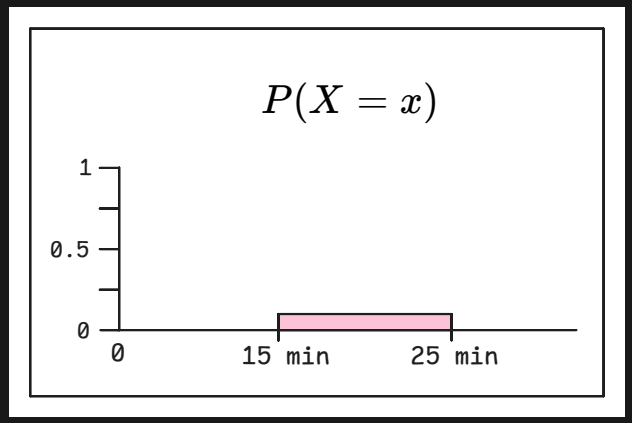

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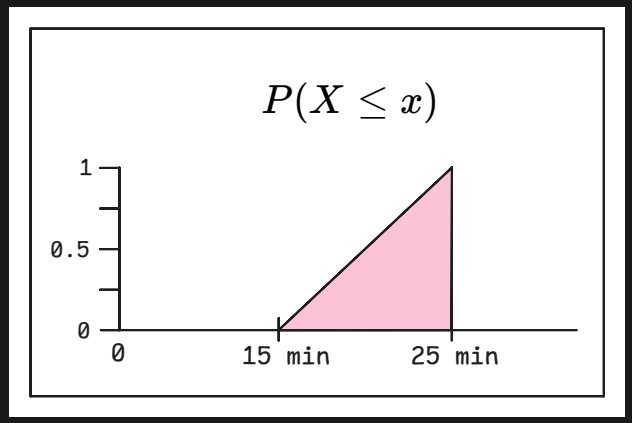

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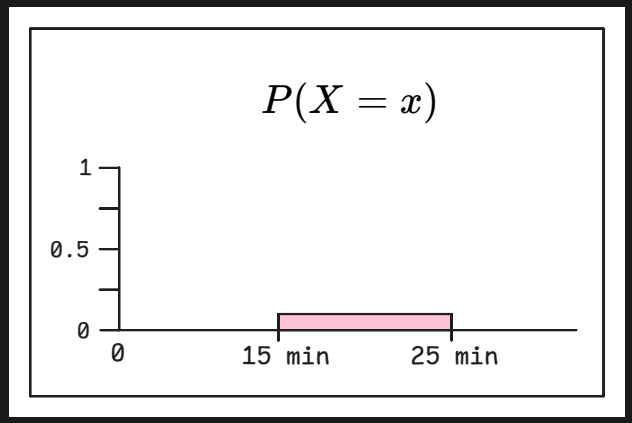

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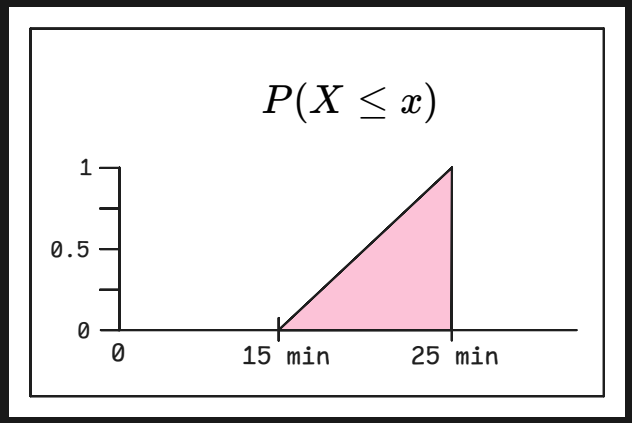

Cumulative distribution function (CDF) $F(x)$ of X

In general:

PDF $f_X(x)$ of a continuous RV X is a non-negative function with total area 1.

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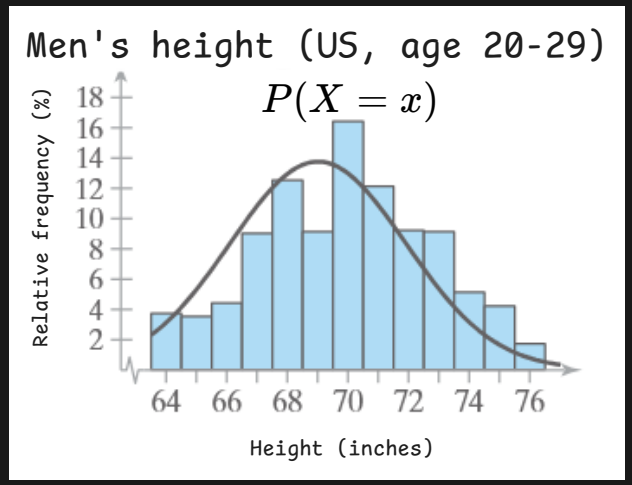
Uniform RV $X \sim U(a, b)$ between a and b has PDF $f_X(x) = \dfrac{1}{b-a}$ if $a \leq x \leq b$ and $f_X(x) = 0$ otherwise.

Normal RV $X \sim N(\mu, \sigma)$:

Let X be the heights of adult male. Then X is approximately normally distributed with $\mu = 70$ in. and $\sigma = 3$ in. Then

$$P(67 \leq X \leq 73) \approx \qquad .$$



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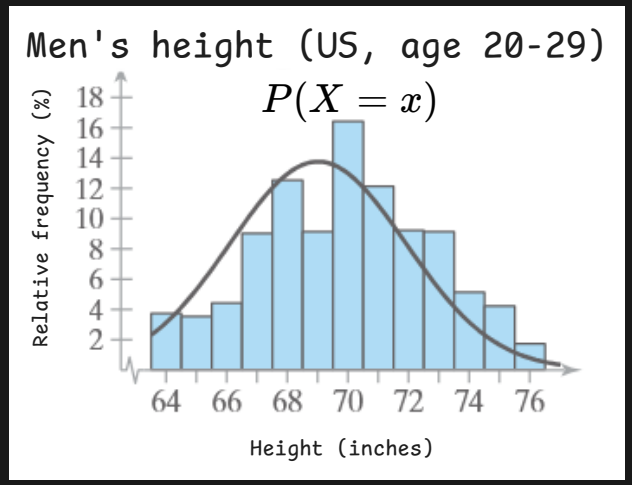

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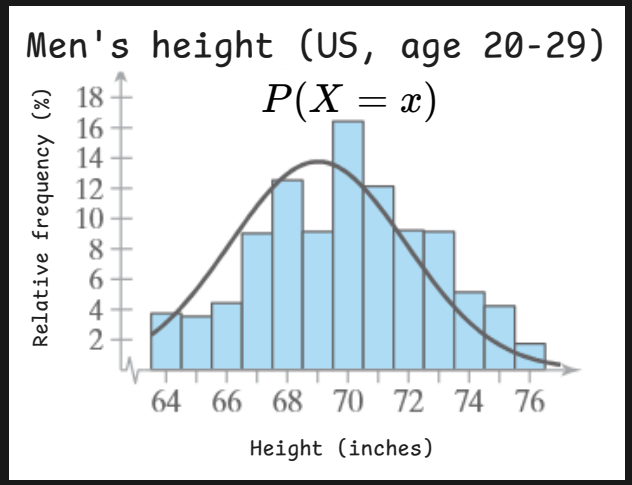

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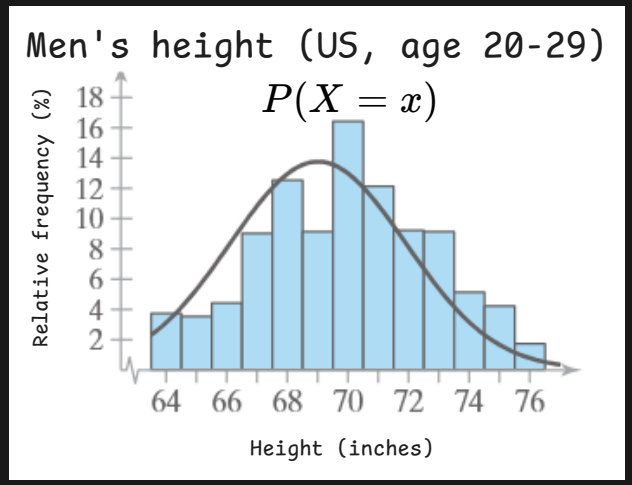
$$P(67 \leq X \leq 73) \approx 0.68.$$

$$F_X(70) \approx 0.50.$$



Probability density function (PDF) $f(x)$ of X

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(c) $\dfrac{1}{2} = F(x) = \sqrt{x} \Rightarrow x = \dfrac{1}{4}$.

In-class exercise

P6. Consider the Pareto random variable X whose PDF has the form c/x^2 for $x \geq 1$ and 0 otherwise.
(a) Find c. (b) Find the CDF of X. (c) Find the median of X.

P2. The proportion of people who respond to a mail-order solicitation is a continuous random variable X that has the density function $f(x) = 2(x+2)/5$ for $0 < x < 1$ and $f(x) = 0$ elsewhere. (a) Show that $P(0 < x < 1) = 1$. (b) Find the probability that more than 1/4 but fewer than 3/4 of the people contacted will respond to this type of solicitation.